

# Compensation of Axial Vibrations in Ball Screw Drives

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## Abstract

This paper presents a precision control strategy for ball screw drives. Axial vibrations are modeled and actively compensated in the control law, which enables the realization of high positioning bandwidth. Lead errors, arising from imperfections of the screw, are modeled and removed from the loop by offsetting their effect from the command trajectory and position feedback signals. Effectiveness of the proposed strategy is demonstrated in high speed tracking experiments conducted on a ball screw drive, where a linear positioning accuracy of 2.6  $\mu\text{m}$  has been maintained while traversing the axis at 1000 mm/s feed with 0.5 g acceleration.

## Keywords:

Feed, Drive, Control

## 1 INTRODUCTION

Demands for increased productivity and higher part quality have motivated a significant amount of research in the past two decades, dedicated to developing faster and more accurate feed drive systems. Encompassed in these efforts were the development of new control strategies [1], [2], [3], [4] to enhance the dynamic accuracy of feed drives, and smooth trajectory generation techniques [5], [6], to shape the motion commands in accordance with the drives' physical capabilities. Advances in these two areas have played complementary roles in achieving accurate multi axis feed motion in high speed machine tools.

One of the most important factors that affect the high speed positioning accuracy is the closed loop bandwidth [7]. In ball screw drives, structural vibrations originating from the torsional and axial modes have a detrimental effect on the achievable bandwidth [8], [9], [10], [11]. In addition, these vibrations also deteriorate the positioning accuracy, and therefore the machined part quality. Various approaches have been proposed in order to deal with structural vibrations in feed drives. Most of these techniques consist either of prefiltering the motion commands before applying them to the servo controller [12], [13], or placing notch filters inside the loop, in order to attenuate the control signal in the vicinity of the resonances [14]. The latter approach also enables a reasonable improvement in the control bandwidth. In recent years, newer strategies have emerged which target active damping of structural vibrations inside the control loop [15], [16], [17], [18]. The main advantage of these strategies is that they enable higher control bandwidth to be achieved, in comparison to placing notch filters inside the loop. However, the design of such controllers is more challenging, as higher order plant models need to be considered in order to capture the vibratory dynamics.

Besides the control bandwidth, other factors such as elastic deformations of the drive mechanism, motion loss in the preloaded nut [19], and imperfections of the mechanical components [20] also influence the final accuracy of a drive system. These factors need to be compensated to a practical extent when designing a high performance control law.

This paper presents a new control strategy for achieving high positioning accuracy in ball screw drives. An adaptive Sliding Mode Controller (SMC) is designed in Section 2, which actively dampens out the axial vibrations of the drive. This leads to a substantial improvement in the bandwidth over placing a notch filter inside the loop. The control law also has the ability to counteract external disturbances. In Section 3, the effect of lead error, which deteriorates the high speed positioning accuracy, is modeled and removed from the servo loop. Effectiveness of the overall control strategy is verified in experiments in Section 4. The conclusions are presented in Section 5.

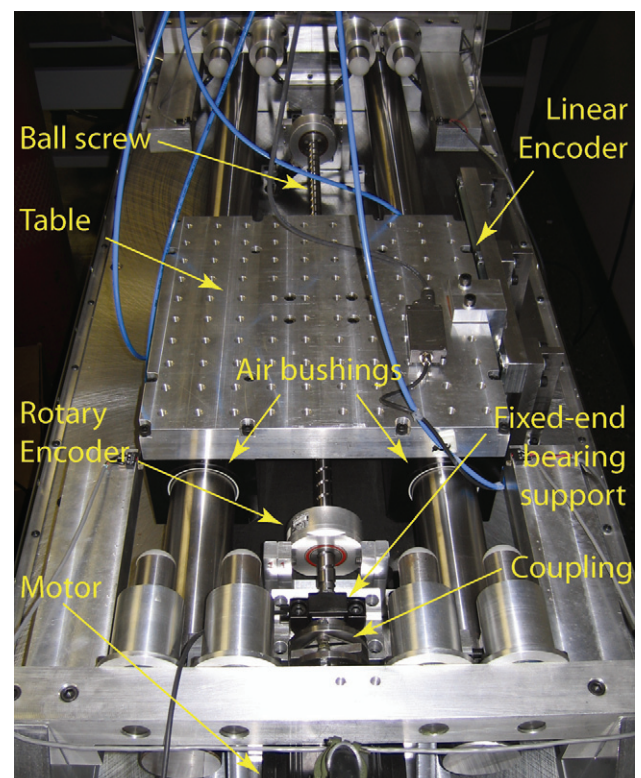


Figure 1: Ball screw drive setup.

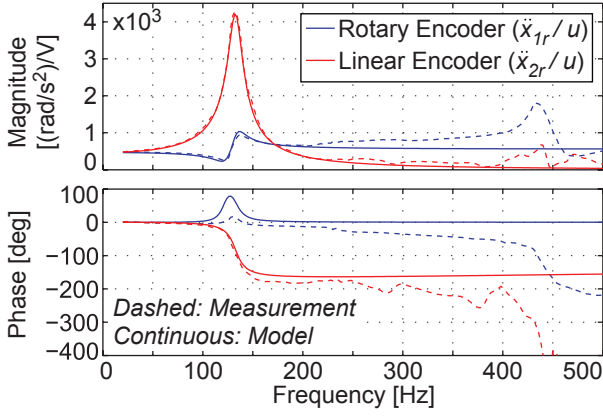


Figure 2: Open loop acceleration FRF's.

## 2 MODE COMPENSATING CONTROLLER DESIGN

### 2.1 Drive Model

The experimental setup used is shown in Figure 1. A precision ball screw with 20 mm pitch and 20 mm diameter drives a table supported on an air guideway system. Actuation is provided through a 3 kW AC servomotor, which is attached to the ball screw using a diaphragm type coupling, in order to avoid rotational backlash. The motor is operated in current control mode which has a bandwidth of 440 Hz. Two feedback devices are used comprising of a rotary encoder mounted on the ball screw, and a linear encoder mounted on the table. Both encoders provide sinusoidal output signals, which result in a position measurement resolution equivalent to 10-20 nm of table motion, when interpolated in the motion controller. The setup is controlled using a dSPACE system, where the developed control law has been implemented at a sampling frequency of 17 kHz.

In earlier work [17] the torsional modes of the ball screw were identified to be at 445, 1080, and 1755 Hz, and a motion controller capable of achieving high rotational accuracy was developed. This paper focuses on modeling and compensation of the axial vibrations, in particular the 1st axial mode, with the goal of enhancing the final linear positioning bandwidth and accuracy.

To identify the 1st axial mode, the open loop Frequency Response Functions (FRF's) relating the motor torque command to the ball screw and table acceleration were measured, as shown in Figure 2. To have consistency in units, the table displacement has been represented in terms of the equivalent ball screw angular motion, by making use of the pitch length ( $h_p = 20 \text{ mm} \Rightarrow 1 \text{ rad} = 3.1831 \text{ mm}$ ). The combined gain for the motor and amplifier is  $K_a K_t = 0.98 \text{ Nm/V}$ . As seen in the figure, the first axial mode is at 133 Hz and is very lightly damped ( $\zeta = 0.056$ ). When the position loop is closed using a linear encoder, and this resonance is not compensated, it brings about a severe limitation in terms of the achievable control bandwidth. The second resonance occurs at around 445 Hz and is attributed to the 1st torsional mode. Attenuation of this mode was studied earlier using active and passive techniques [17], different from the approach presented in this paper.

A two mass drive model, shown in Figure 3, was used in capturing the key dynamics of the ball screw drive including the 1st axial mode. In the figure,  $m_1$  represents the inertia of rotating components, such as the motor, the coupling, and the ball screw.  $m_2$  represents the table inertia, which is subject to axial vibrations.  $b_1$  and  $b_2$

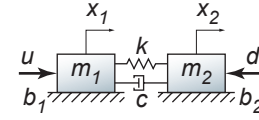


Figure 3: Flexible drive model used in controller design.

represent the viscous friction in the rotary bearings and linear guideways, respectively.  $k$  is the overall axial stiffness and  $c$  represents the damping induced in the preloaded nut.  $u$  is the motor torque command (i.e. control signal) and  $d$  represents external disturbances acting on the table, such as cutting forces and unmodeled nonlinear friction.  $x_1$  and  $x_2$  represent the ball screw rotational and table linear displacements, respectively.

The total inertia and viscous friction were identified using Least Squares. With knowledge of the drive's mass distribution, individual inertia values were calculated to be  $m_1 = 1.858 \times 10^{-3}$ ,  $m_2 = 0.346 \times 10^{-3} \text{ V/(rad/s}^2\text{)}$ .  $b_1 = 1.020 \times 10^{-3}$ , and since the setup employs air guideways  $b_2 = 0 \text{ V/(rad/s)}$ . On machines that have lubricated or roller guides, a nonzero value for  $b_2$  would be identified.

There are several factors which influence the equivalent axial stiffness  $k$ . In particular, these are the individual stiffness values of the ball screw, preloaded nut, and thrust bearing. In addition, the preloaded nut serves as a special spring that couples different degrees of freedom together, especially in the torsional and axial directions. Detailed Finite Element studies have been conducted to analyze and simulate these effects [10], [11], which require knowledge of the individual stiffness values. The damping, on the other hand, is difficult to predict analytically and is best identified by taking measurements from the real machine. In this study, the axial stiffness and damping were identified by matching the frequency response of the model to the measured FRF's, as seen in Figure 2. The resulting values are  $k = 202.21 \text{ V/rad}$  and  $c = 0.027 \text{ V/(rad/s)}$ . It is seen that the model matches the drive's FRF closely up to 250 Hz. Above this frequency, additional dynamics attributed to the current loop and the 1st torsional mode become dominant, causing a variation from the modeled frequency response. The drive model can be written in state space as:

$$\begin{bmatrix} \dot{x}_2 \\ \ddot{x}_2 \\ \dot{x}_1 \\ \ddot{x}_1 \end{bmatrix} = \underbrace{\begin{bmatrix} 0 & 1 & 0 & 0 \\ -k & -c-b_2 & k & c \\ \frac{m_2}{m_1} & \frac{m_2}{m_1} & \frac{m_2}{m_1} & \frac{m_2}{m_1} \\ 0 & 0 & 0 & 1 \end{bmatrix}}_{\mathbf{A}} \underbrace{\begin{bmatrix} x_2 \\ \dot{x}_2 \\ x_1 \\ \dot{x}_1 \end{bmatrix}}_{\mathbf{z}} + \underbrace{\begin{bmatrix} 0 \\ 0 \\ 0 \\ \frac{1}{m_1} \end{bmatrix}}_{\mathbf{B}} u \quad (1)$$

$$+ \underbrace{\begin{bmatrix} 0 \\ -1 \\ \frac{1}{m_2} \\ 0 \\ 0 \end{bmatrix}}_{\mathbf{W}} \underbrace{\begin{bmatrix} d \end{bmatrix}}_{\mathbf{d}} = \underbrace{\begin{bmatrix} \mathbf{A}_{11} & \mathbf{A}_{12} \\ \mathbf{A}_{21} & \mathbf{A}_{22} \end{bmatrix}}_{\mathbf{A}} \underbrace{\begin{bmatrix} \mathbf{z}_1 \\ \mathbf{z}_2 \end{bmatrix}}_{\mathbf{z}} + \underbrace{\begin{bmatrix} \mathbf{B}_1 \\ \mathbf{B}_2 \end{bmatrix}}_{\mathbf{B}} u + \underbrace{\begin{bmatrix} \mathbf{W}_1 \\ \mathbf{W}_2 \end{bmatrix}}_{\mathbf{W}} \mathbf{d}$$

Above, the state vector  $\mathbf{z}$  is divided into two parts.  $\mathbf{z}_2 = [\dot{x}_1]$  (velocity of  $m_1$ ) is directly influenced by the input torque ( $\mathbf{B}_2 = [1/m_1]$ ), whereas the remaining states  $\mathbf{z}_1 = [x_2 \ \dot{x}_2 \ x_1]^T$  are influenced indirectly ( $\mathbf{B}_1 = \mathbf{0}_{3 \times 1}$ ), through the  $\mathbf{A}_{12}\mathbf{z}_2$  term. This separation is essential for formulating the sliding surface in the control law design.

## 2.2 Sliding Surface Design

Sliding surface selection is the first step in designing an adaptive Sliding Mode Controller (SMC). In this work, this is achieved using the Method of Equivalent Control [21], as explained in the following. Considering at first only the regulation problem for the states in  $\mathbf{z}_1$  and ignoring the effect of external disturbances (i.e.  $\mathbf{Wd} = \mathbf{0}$ ), the objective is to constrain the system states on a stable sliding surface so that they eventually converge to zero. The sliding surface is defined as:

$$\sigma = [s_1 \ s_2 \ s_3 \ s_4] \cdot \begin{bmatrix} x_2 \\ \dot{x}_2 \\ x_1 \\ \dot{x}_1 \end{bmatrix} = \mathbf{S}\mathbf{z} = \mathbf{S}_1\mathbf{z}_1 + \mathbf{S}_2\mathbf{z}_2 = 0 \quad (2)$$

where  $\mathbf{S}_1 = [s_1 \ s_2 \ s_3]$  and  $\mathbf{S}_2 = [s_4]$ . When the states are on the sliding surface  $\sigma = 0$  should hold, or in other words the system states should satisfy:  $\mathbf{z}_2 = -\mathbf{S}_2^{-1}\mathbf{S}_1\mathbf{z}_1$ . Substituting this into the state transition expression for  $\dot{\mathbf{z}}_1$  (Equation 1) results in:

$$\dot{\mathbf{z}}_1 = \mathbf{A}_{11}\mathbf{z}_1 + \mathbf{A}_{12}\mathbf{z}_2 = [\mathbf{A}_{11} - \mathbf{A}_{12}(\mathbf{S}_2^{-1}\mathbf{S}_1)] \cdot \mathbf{z}_1 \quad (3)$$

The sliding surface parameters  $s_1, s_2, s_3, s_4$  need to be determined so that the closed loop poles (i.e. eigenvalues of  $[\mathbf{A}_{11} - \mathbf{A}_{12}(\mathbf{S}_2^{-1}\mathbf{S}_1)]$ ) can be assigned the desired natural frequency ( $\lambda, \omega_n$ ) and damping ( $\zeta$ ) values:

$$\text{eig}[\mathbf{A}_{11} - \mathbf{A}_{12}(\mathbf{S}_2^{-1}\mathbf{S}_1)] = \begin{cases} -\lambda \\ -\zeta\omega_n \pm j\omega_n\sqrt{1-\zeta^2} \end{cases} \quad (4)$$

This is achieved using the standard pole placement design technique, by considering the similarity between Equation 3 and the state feedback gain selection problem stated as: *Find  $\mathbf{K}$  so that  $\dot{\mathbf{x}} = (\mathbf{A} - \mathbf{BK})\mathbf{x}$  has the desired eigenvalues.* Since there are 3 pole locations and 4 sliding surface parameters, the designer has to make a choice for one of the values in  $\mathbf{S}$ , which does not really affect the outcome. In this work,  $s_4$  has been chosen as “1” for convenience. Since our design deals with trajectory tracking, the sliding surface is modified to consider the state tracking errors as,

$$\sigma = \mathbf{S}(\mathbf{z}_r - \mathbf{z}) = \underbrace{[s_1 \ s_2 \ s_3 \ 1]}_{\mathbf{S}} \cdot \left\{ \begin{bmatrix} x_{2r} \\ \dot{x}_{2r} \\ x_{1r} \\ \dot{x}_{1r} \end{bmatrix} - \begin{bmatrix} x_2 \\ \dot{x}_2 \\ x_1 \\ \dot{x}_1 \end{bmatrix} \right\} \quad (5)$$

Above,  $\mathbf{z}_r$  represents the reference state, comprising of position and velocity commands for  $m_2$  and  $m_1$ .

## 2.3 Control Law

The sliding mode controller is designed by following the procedure set forth in [22] and [23]. Further details of the technique and stability proof can be found in the cited papers. Here, the design is adapted to achieve simultaneous position and vibration control in ball screw drives. The external disturbance  $d$  is considered as the only unknown parameter, which needs to be adapted and cancelled out in the control law. The controller design is

carried out by constructing the parameter-regressor expression in the form:

$$(\mathbf{SB})^{-1}\mathbf{S}(\dot{\mathbf{z}}_r - \mathbf{A}\mathbf{z} - \mathbf{Wd}) = \mathbf{Y}(\dot{\mathbf{z}}_r, \mathbf{z}) \cdot \mathbf{P} \quad (6)$$

Above,  $\mathbf{P}$  contains the unknown parameters and  $\mathbf{Y}$  contains the corresponding regressors. Using the drive dynamics in Equation 1 and the sliding surface definition in Equation 5, the parameter-regressor expression is obtained as,

$$\mathbf{Y}(\dot{\mathbf{z}}_r, \mathbf{z}) \cdot \mathbf{P} = [y_1 \ y_2] \cdot \begin{bmatrix} 1 \\ d \end{bmatrix} = y_1 + y_2 d \quad (7)$$

where,

$$\left. \begin{aligned} y_1 &= m_1 s_1 (\dot{x}_{2r} - \dot{x}_2) + m_1 s_3 (\dot{x}_{1r} - \dot{x}_1) \\ &\quad + m_1 s_2 (\ddot{x}_{2r} + \frac{k}{m_2} x_2 + \frac{c+b_2}{m_2} \dot{x}_2 - \frac{k}{m_2} x_1 - \frac{c}{m_2} \dot{x}_1) \\ &\quad + m_1 (\ddot{x}_{1r} - \frac{k}{m_1} x_2 - \frac{c}{m_1} \dot{x}_2 + \frac{k}{m_1} x_1 + \frac{c+b_1}{m_1} \dot{x}_1) \\ y_2 &= (m_1 / m_2) s_2 \end{aligned} \right\} \quad (8)$$

Considering a Lyapunov function postulated as:

$$V = \frac{1}{2} [\sigma^T (\mathbf{SB})^{-1} \sigma + (\mathbf{P} - \hat{\mathbf{P}})^T \boldsymbol{\Gamma} (\mathbf{P} - \hat{\mathbf{P}})] \quad (9)$$

where  $\mathbf{SB}$  is a positive definite matrix and  $\boldsymbol{\Gamma}$  is a diagonal matrix with positive entries, this function will be positive when the states are not on the sliding surface ( $\sigma \neq 0$ ) or there is discrepancy between the actual and estimated system parameters (i.e.  $\mathbf{P} \neq \hat{\mathbf{P}}$ ). It was shown in [23] that as long as the real and estimated parameters are within known bounds (i.e.  $\mathbf{P}_{\min} \leq \mathbf{P} \leq \mathbf{P}_{\max}$  and  $\mathbf{P}_{\min} \leq \hat{\mathbf{P}} \leq \mathbf{P}_{\max}$ ), a control law to stabilize the system by achieving  $\dot{V} < 0$  can be obtained as:

$$u_{smc} = K_s \sigma + \mathbf{Y}(\dot{\mathbf{z}}_r, \mathbf{z}) \cdot \hat{\mathbf{P}} = K_s \sigma + y_1 + y_2 \hat{d} \quad (10)$$

In the control law,  $K_s$  is a positive feedback gain to push the state errors on to the sliding surface. Expressions for  $\sigma$ ,  $y_1$ , and  $y_2$  were given in Equations 5 and 8. In the ideal case, where the actual drive dynamics perfectly matches the model in Equation 1, the disturbance adaptation would be implemented as [22], [23]:

$$\dot{\hat{d}} = y_2 \rho \kappa \sigma = \frac{m_1 s_2}{m_2} \rho \kappa \left[ s_1 (x_{2r} - x_2) + s_2 (\dot{x}_{2r} - \dot{x}_2) + s_3 (x_{1r} - x_1) + \dot{x}_{1r} - \dot{x}_1 \right] \quad (11)$$

Above,  $\rho > 0$  is the adaptation gain and  $\kappa$  is a binary variable to switch the adaptation on and off in order to contain the disturbance estimate within specified bounds ( $d_{\min} \leq \hat{d} \leq d_{\max}$ ) [4]. In experimental trials, it was seen that this adaptation law, which uses feedback from both rotary and linear encoders, was not successful in achieving zero steady state error in the final linear position. This is attributed to imperfections in the drive's rotary to linear motion delivery, which exhibits discrepancies in the order of a few microns from the ideal model in Equation 1. In order to alleviate this problem, a simpler disturbance adaptation has been applied, similar to the one used in [4] and [17]. This adaptation law uses a first order sliding surface derived considering only rigid body dynamics, observed through the linear encoder

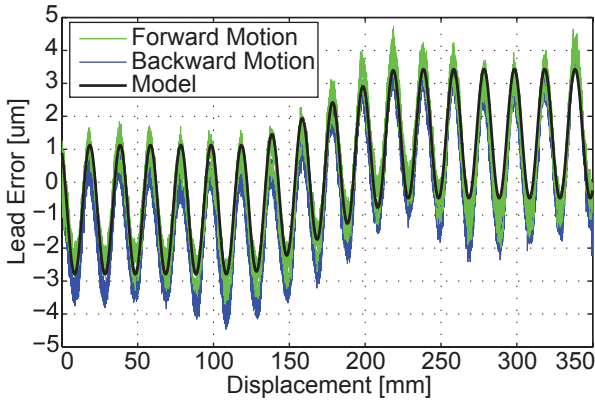


Figure 4: Measured and modeled lead error profiles.

( $\sigma' = \dot{x}_{2r} - \dot{x}_2 + \lambda(x_{2r} - x_2)$ ). Applying this adaptation law produced much better results in terms of the final linear positioning accuracy:

$$\hat{d} = y_2 \rho K \sigma' = \frac{m_1 s_2}{m_2} \rho K [\dot{x}_{2r} - \dot{x}_2 + \lambda(x_{2r} - x_2)] \quad (12)$$

## 2.4 State Trajectory Generation

One final requirement is the generation of state command trajectories. Since the primary objective is to have the table position ( $x_2$ ) track the given motion commands ( $x_R$ ),  $x_{2r}$  is chosen as:

$$x_{2r} = x_R \quad (13)$$

On the other hand, the ball screw rotary position ( $x_1$ ) has to be offset in order to account for the axial deformations that occur during accelerations and decelerations of the drive, or when external forces are applied on the table. Writing the equation of motion for  $m_2$ ,

$$m_2 \ddot{x}_2 = k(x_1 - x_2) + c(\dot{x}_1 - \dot{x}_2) - b_2 \dot{x}_2 - d \quad (14)$$

and assuming that  $x_2 \equiv x_{2r}$  and  $d \equiv \hat{d}$ , the position command for  $m_1$  is obtained as:

$$x_{1r} = \frac{1}{cs + k} \left[ m_2 \underbrace{s^2 x_{2r}}_{\ddot{x}_{2r}} + (c + b_2) \underbrace{s x_{2r}}_{\dot{x}_{2r}} + k x_{2r} + \hat{d} \right] \quad (15)$$

It should be noted that the control law in Equation 10 requires the computation of acceleration commands  $\ddot{x}_{1r}$  and  $\ddot{x}_{2r}$  while constructing the  $y_1$  term given by Equation 8. Estimation of  $\ddot{x}_{1r}$  requires the original command trajectory  $x_R$  to be differentiated four times with respect to time and then low-pass filtered at a cut-off frequency of  $k/c$  (Equation 15). It is therefore necessary that a jerk continuous trajectory similar to the one in [17] be used, in order to avoid delivering a discontinuous control signal that would excite the structural modes of the drive. In digital implementation, the numerical differentiation and integration were carried out using Euler's approximation technique ( $s \equiv (z-1)/(T_s z)$ ). A sampling frequency of 17 kHz was used, which was found to be sufficient in achieving successful vibration control on the ball screw drive.

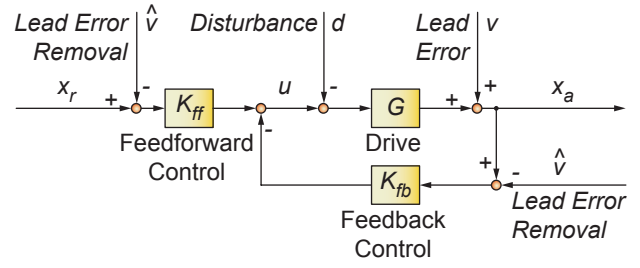


Figure 5: Removal of lead errors from the control loop.

## 3 COMPENSATION OF LEAD ERRORS

Using linear feedback, it is possible to detect and eliminate the positioning errors that originate from imperfections of the drive mechanism, in steady state. The lead error is a good example of this. However, when the axis is in motion, the lead error takes the form of a sinusoidal disturbance that continuously offsets the drive's true position. At high speeds, the control law cannot immediately detect and compensate for this error, which results in the deterioration of tracking accuracy. In order to alleviate this problem, a model based compensation strategy has been used. The lead errors were measured by moving the axis back and forth at low speeds ( $\sim 10$  mm/s), to minimize elastic deformations, and measuring the difference between linear and rotary encoder readings. Measurement for 10 back and forth movements is shown in Figure 4. It was seen that these errors remained reasonably consistent when measurements were taken at different times under similar conditions. Defining the ideal and actual linear positions as  $x_2$  and  $x_{2a}$  respectively, the lead error  $v$  is expressed as:

$$v = x_{2a} - x_2 \quad (16)$$

For the setup, the below model was found to be adequate:

$$\hat{v}(x) = A \sin\left(\frac{2\pi}{h_p} x + \phi\right) + mx + n \quad (17)$$

Above,  $\hat{v}$  (mm) is the modeled lead error and  $x$  (mm) is the nominal axis position. Using Least Squares, the model parameters were identified as  $A = 1.960 \times 10^{-3}$  mm,  $\phi = 2.0627$  rad,  $h_p = 20$  mm. In the range  $x < 125$  mm,  $m = 0$  and  $n = -0.833 \times 10^{-3}$  mm. For  $125 \leq x < 220$  mm,  $m = 2.4262 \times 10^{-5}$ ,  $n = -3.866 \times 10^{-3}$  mm. When  $x \geq 220$ ,  $m = 0$ ,  $n = 1.472 \times 10^{-3}$  mm. While implementing this model in the control law, the displacement units were changed to radians, in order to remain consistent with the controller formulation developed in Section 2.

The concept of lead error compensation is illustrated with a simple example in Figure 5, where the effect of lead error  $v$  is removed from the loop by subtracting its predicted value  $\hat{v}$  from the commanded and measured axis positions ( $x_r$  and  $x_a$  respectively). In this case, the final axis position  $x_a$  can be expressed as:

$$x_a = \underbrace{\frac{GK_{ff}}{1+GK_{fb}} \cdot x_r}_{\text{Tracking}} - \underbrace{\frac{G}{1+GK_{fb}} \cdot d}_{\text{Disturbance}} + \underbrace{\frac{1}{1+GK_{fb}} \cdot v}_{\text{Lead Error Transmission}} - \underbrace{\frac{G(K_{ff} - K_{fb})}{1+GK_{fb}} \cdot \hat{v}}_{\text{Lead Error Compensation}} \quad (18)$$

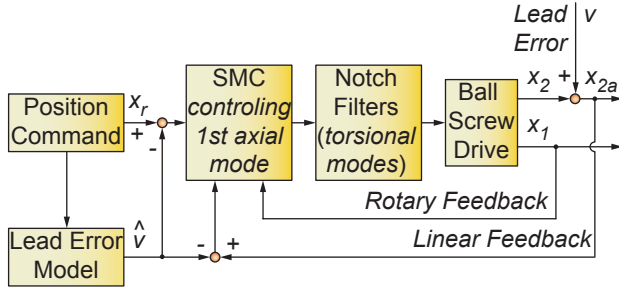


Figure 6: Proposed control structure.

In a well designed controller, the tracking transfer function will be close to unity in the frequency range of most motion commands (i.e.  $GK_{ff} \approx 1 + GK_{fb}$ ). Assuming that the identified model represents the actual lead error with sufficient closeness (i.e.  $\hat{v} \approx v$ ), it can be verified that the compensation term in Equation 18 will cancel out the effect of lead errors transmitted into the servo loop, thus mitigating their effect on the dynamic performance of the drive. It can also be verified that this compensation strategy will produce the same outcome with the dual feedback control law developed in Section 2, which is not derived here for conciseness.

The overall control scheme is shown in Figure 6. Control of rigid body motion and the 1st axial vibration mode is achieved using adaptive Sliding Mode Control. The sliding surface was designed to yield high tracking accuracy and good disturbance rejection, while attenuating vibrations of the 1st axial mode as much as possible and avoiding the excitation of unmodeled dynamics at higher frequencies. Following trials on the experimental setup, it was seen that placing the real pole at  $\lambda = 1131$  rad/s (180 Hz), and the complex conjugate poles at  $\omega_n = 628$  rad/s (100 Hz) with  $\zeta = 0.35$  damping realized these goals. Although the damping is not very high, it is six times higher than the open loop damping of 5.6%, and is achieved in closed loop. It was observed that further increasing the pole frequencies or damping caused the control signal to become aggressive at high frequencies, which led to undesirable vibrations. The feedback and disturbance adaptation gains were tuned to be  $K_s = 0.005$  V/(rad/s) and  $\rho = 7.751$  V/rad.

Lead errors were compensated using the model in Equation 17. Considering that this model has first order discontinuity at transition points  $x = 125$  and  $220$  mm, the compensation signal was low pass filtered at 250 Hz, in order to mitigate the effect of this discontinuity on the drive's performance. Notch filters were also incorporated into the control scheme, in order to avoid excitation of the 1st and 2nd torsional resonances [17], which made vibration control of the 1st axial mode possible.

#### 4 EXPERIMENTAL RESULTS

The proposed control scheme has been evaluated in frequency response and high speed tracking experiments conducted on the ball screw drive. In frequency response tests, the mode compensating controller has been compared to an alternative design which consists of using a notch filter to attenuate the 1st axial mode and considering only rigid body dynamics in designing the SMC, similar to the approach in [17]. The loop was closed using linear encoder feedback and additional notch filters were inserted to attenuate the 1st and 2nd torsional modes. In this case, the sliding surface bandwidth could be increased only up to  $\lambda = 440$  rad/s (70 Hz). The sampling frequency for this controller was 17 kHz.

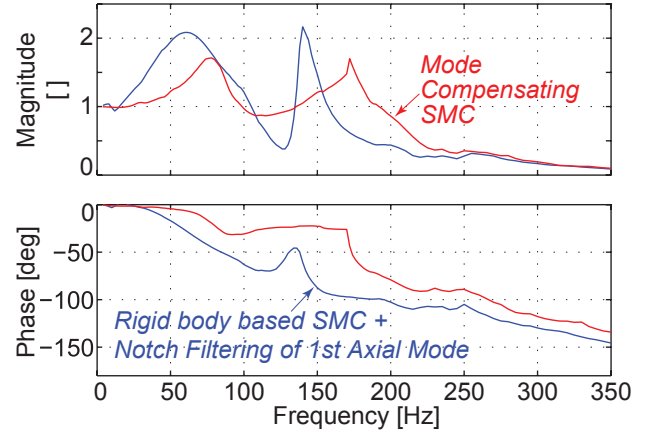


Figure 7: Experimental FRF's for linear positioning.

Experimentally measured FRF's comparing the tracking performance of the two controllers are shown in Figure 7. It can be seen that the mode compensating controller has wider usable frequency range, where the gain drops below -3 db after 208 Hz. The -3 db drop occurs at 113 Hz for the rigid body based SMC, which indicates a two fold bandwidth difference. In addition, the gain of the mode compensating SMC is closer to unity and its phase lag is considerably lower (less than 30 deg) up to 170 Hz, indicating a more favorable tracking performance.

This performance improvement was verified in high speed tracking experiments. For a jerk continuous trajectory with 350 mm displacement, 1000 mm/s feed, 5000 mm/s<sup>2</sup> acceleration, and 100,000 mm/s<sup>3</sup> jerk, it was seen that the rigid body based SMC with notch filtering could achieve a linear tracking accuracy of 21  $\mu$ m. By tuning this controller further, it was possible to bring the tracking error down to 9.2  $\mu$ m. However, this over tuning resulted in sharp resonances in the closed loop FRF with magnitudes reaching as high as 4.6, which was highly undesirable. For conciseness, tracking results obtained with this controller are not presented here.

The tracking performance of the mode compensating SMC was evaluated without and with lead error compensation. Results of these tests are shown in Figures 8 and 9. When lead error compensation is not used, a final linear accuracy of 6.5  $\mu$ m is achieved, as seen in Figure 8. This is a better accuracy than that achieved with rigid body based SMC and notch filtering of the 1st axial mode. Furthermore, the closed loop frequency response is not distorted due to over tuning. The effect of lead errors can clearly be seen in the tracking error profile, as if a sinusoidal disturbance is applied at the drive's output. When lead error compensation is enabled (Figure 9), the tracking error is further reduced to 2.6  $\mu$ m, indicating an overall improvement in the drive's linear positioning accuracy, achieved with the proposed controller design.

#### 5 CONCLUSIONS

This paper has presented a new control strategy for achieving high positioning accuracy in ball screw drives. The position loop is closed using adaptive sliding mode control, which is designed to actively attenuate vibrations of the 1st axial mode. It is shown that this controller is capable of achieving higher positioning bandwidth in comparison to placing a notch filter inside the loop. Lead errors, which are repeatable, are also modeled and compensated in the control law. With the proposed technique, a positioning bandwidth of 208 Hz has been achieved on a ball screw drive, leading to a linear tracking accuracy of 2.6  $\mu$ m while traversing the axis at 1000 mm/s feed with 0.5 g acceleration.

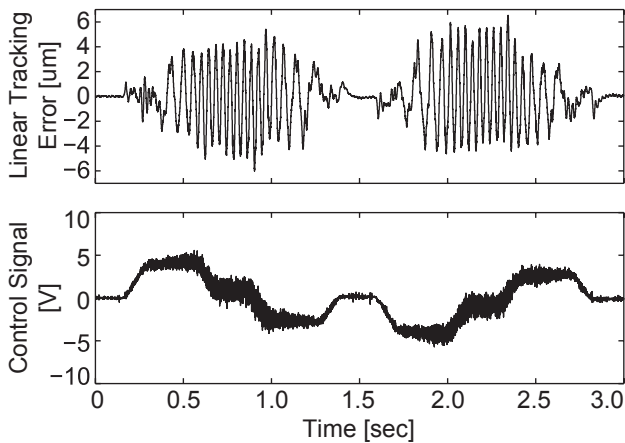


Figure 8: Tracking performance of mode compensating SMC without lead error correction.

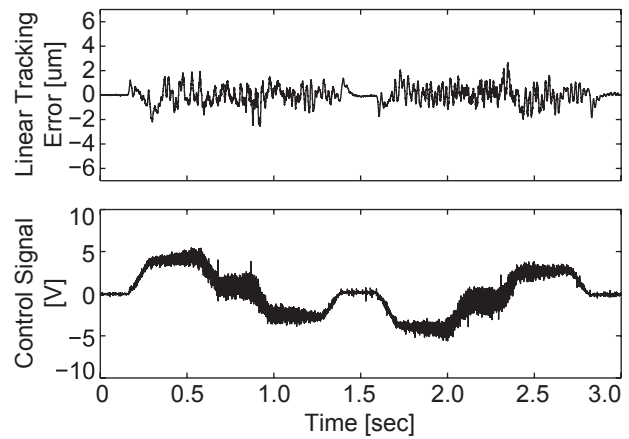


Figure 9: Tracking performance of mode compensating SMC with lead error correction.

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